



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-	Team No-	Project Title- QueueLess – AI-Based Smart Virtual Queue and Waiting-Time Prediction System
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1. Abstract

(Summarize the project idea, problem background, and proposed solution. Not more than 250 words.)

Long waiting lines at hospitals, banks, government offices, and service centers cause frustration, wasted time, and inefficient use of staff resources. Conventional token-based queue systems give customers no visibility into how long they must actually wait, forcing them to stay physically present in line. QueueLess is an AI-based smart virtual queue and waiting-time prediction system designed to solve this problem. The system allows users to join a queue remotely through a web or mobile interface, removing the need to wait physically at the location. It uses machine learning models trained on historical service-time data, staff availability, and real-time queue length to accurately predict individual waiting times and dynamically update them as conditions change. A backend database engine manages queue state, customer records, and service logs in real time, ensuring consistency even under high concurrent load. Customers receive live notifications as their turn approaches, while administrators get a dashboard to monitor queue analytics, staff performance, and peak-hour trends. By combining distributed backend architecture with predictive analytics, QueueLess reduces perceived and actual waiting time, improves customer satisfaction, and helps organizations optimize resource allocation and staffing decisions.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030559	M. Adith Kumar	
2520030307	N.V. Karthikeya	
2520030571	A. Revanth	

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Frontend: React.js; Backend: Node.js / Express.js; Database: MySQL / MongoDB; Machine Learning: Python, Scikit-learn / TensorFlow for wait-time prediction; Real-time Communication: WebSockets / Firebase Cloud Messaging for live notifications; Hosting: AWS / Firebase; Version Control: Git and GitHub.

Guide Name and Signature (remarks if any) _____